

SYDNEY TECHNICAL HIGH SCHOOL

MATHEMATICS EXTENSION II

HSC ASSESSMENT TASK II

JUNE 2002

Time allowed: 70 minutes

Instructions:

- ☐ Show all necessary working in every question.
- ☐ Start each question on a new page.
- ☐ Attempt all questions.
- ☐ All questions are of equal value.
- ☐ Full marks may not be awarded for careless or badly arranged work.
- ☐ Approved calculators may be used.
- ☐ This test forms part of your HSC assessment.
- ☐ These questions are to be handed in with your answers.

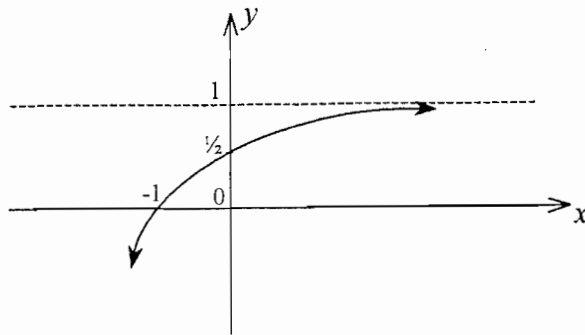
Name: _____

Question 1	Question 2	Question 3	Total

Question 1

Marks

- a) Find $\int \frac{dx}{x^2 - 4x + 13}$ 2
- b) The diagram shows the graph of the increasing function $y = f(x)$ 6



Draw separate one-third page sketches of the graphs of the following showing any intercepts

- i) $y = \frac{1}{f(x)}$
- ii) the inverse function of $y = f(x)$
- iii) $y = \ln f(x)$
- c) Find $\int \frac{dx}{\sqrt{e^{2x} - 1}}$ using the substitution $u = \sqrt{e^{2x} - 1}$ 2
- d) A (1,4) is a fixed point while $P(2t, \frac{2}{t})$ is a variable point in the first quadrant

on the hyperbola $xy = 4$.

7

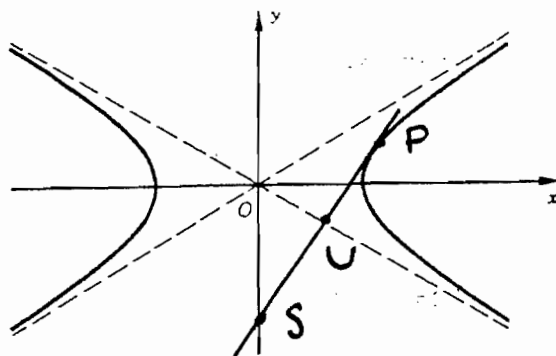
- i) what values can "t" take
- ii) a) Show that the equation of the chord AP is $2x + ty - 4t - 2 = 0$
b) Which value/s of "t" must be excluded why?
- iii) The chord AP cuts the x -axis at M. Find the coordinates of M
- iv) Find the equation of the locus of the mid point of PM.

Question 2

Marks

- a) The polynomial $x^3 + 5x - t = 0$ has 3 real roots α, β, γ 4
- i) Find the cubic equation with roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$
- ii) Hence or otherwise find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$
- b) i) If $\frac{a}{x^2 - a^2} \equiv \frac{A}{x - a} + \frac{B}{x + a}$ find the values of A and B 5
- ii) Use the substitution $x = u^2$ to find $\int \frac{\sqrt{x}}{x-1} dx$
(you may use the result in part i)

c) 8



Consider the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$

- i) Write down the equation of each asymptote
- ii) By differentiation find the gradient of the tangent to the hyperbola at $P(3 \sec \theta, 4 \tan \theta)$
- iii) Show that the equation of the tangent at P is $4x = 3 \sin \theta y + 12 \cos \theta$
- iv) Find the x-coordinate of U the point where the tangent meets the asymptote (as shown on the diagram)
- v) Using the x-values only, find the value for θ such that U is the mid point of PS.

Question 3

- a) Find $\int \tan^3 x \, dx$ 2
- b) Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{2 - 2 \cos 2x} \, dx$ 2
- c) i) Write down the expansion for $\cos 3\theta$ in terms of $\cos \theta$ 5
ii) By solving an equation of the form $\cos 3\theta = K$ and using part i) find the exact roots of the equation $8x^3 - 6x - \sqrt{3} = 0$
- d) i) Explain why a polynomial of degree 5 with real co-efficients has at least 1 real root. 7

The polynomial $P(x)$ is given by $P(x) = x^5 - 5cx + 1$ where c is a real number

- ii) Show by considering the turning points of $P(x)$, that when $c < 0$, $P(x)$ has just one real root.
- iii) Explain why this root is negative
- iv) Prove that $P(x)$ has three distinct real roots if $c > \left(\frac{1}{4}\right)^{\frac{1}{5}}$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

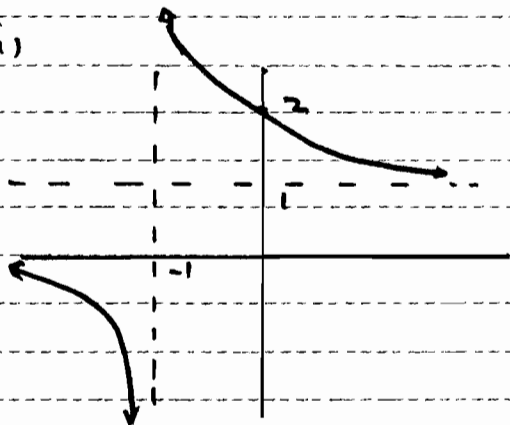
NOTE: $\ln x = \log_e x, \quad x > 0$

①

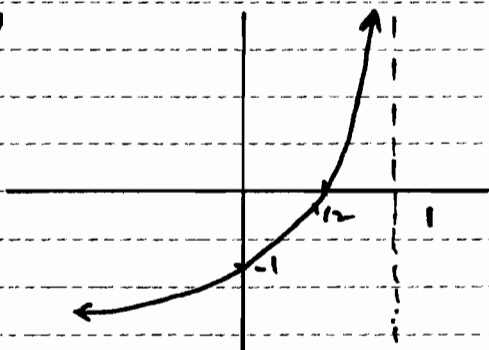
a)

$$\begin{aligned}\int \frac{dx}{x^2 - 4x + 13} &= \int \frac{dx}{(x^2 - 4x + 4) + 9} \\ &= \int \frac{dx}{9 + (x-2)^2} \checkmark \\ &= \frac{1}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + c \checkmark\end{aligned}$$

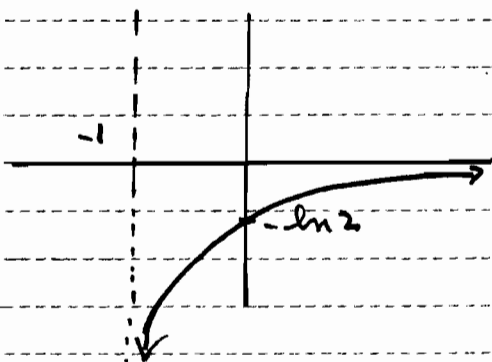
b) i)



ii)



iii)



each 1 shape.

, intercepts/asymptotes

c)

$$\begin{aligned}u^2 &= e^{2x} - 1 \\ 2u du &= 2e^{2x} dx \\ dx &= \frac{u du}{e^{2x}} \checkmark\end{aligned}$$

$$\therefore \int \frac{u du}{(u^2 + 1) \cdot u}$$

$$= \int \frac{du}{u^2 + 1}$$

$$= \tan^{-1} \sqrt{e^{2x} - 1} \checkmark$$

d) i) $t > 0$ ✓

$$\text{ii) c) } mAP = \frac{2-4}{2t-1}$$

$$= \frac{2-4t}{t(2t-1)}$$

$$= \frac{-2(2t-1)}{t(2t-1)}$$

$$= -\frac{2}{t} \checkmark$$

$$y-4 = -\frac{2}{t}(x-1)$$

$$ty-4t = -2x+2$$

$$2x+ty-4t-2=0 \checkmark$$

i) b) $t \neq 1/2$ Coincides with A ✓iii) $y=0$

$$2x = 4t+2$$

$$x = 2t+1$$

$$\therefore M(2t+1, 0) \checkmark$$

iv) mid pt PM

$$= \frac{2t+2t+1}{2}, \frac{0+2}{2}$$

$$= \left(\frac{4t+1}{2}, \frac{1}{t} \right) \checkmark$$

 $\therefore$ Locus

$$t = 1/y \text{ sub in } x = \frac{4t+1}{2}$$

a) i) Let $y = \frac{1}{x^2}$

$\therefore x = \frac{1}{\sqrt{y}}$

x is a root of $x^3 + 5x - t = 0$

$\therefore \left(\frac{1}{\sqrt{y}}\right)^3 + 5\left(\frac{1}{\sqrt{y}}\right) - t = 0$

$\frac{1}{y\sqrt{y}} + \frac{5}{\sqrt{y}} - t = 0$

$1 + 5y - ty\sqrt{y} = 0$

$1 + 5y = ty\sqrt{y}$

s.b.s

$1 + 10y^2 + 25y^2 = t^2y^3$

$\therefore t^2y^3 - 25y^2 - 10y^2 - 1 = 0$

ii) $\frac{1}{x^2} = \frac{25}{t^2}$

b) i) $\frac{a}{x^2 - a^2} = \frac{A}{x - a} + \frac{B}{x + a}$

$A = \frac{1}{2}, B = -\frac{1}{2}$

ii) $\int \frac{\sqrt{x}}{x-1} dx$

$x = u^2$

$dx = 2u du$

$\int \frac{u \cdot 2u du}{u^2 - 1}$

$\int \frac{2u^2 du}{u^2 - 1}$

$2 \int \frac{u^2 - 1 + 1}{u^2 - 1} \cdot \frac{1}{u^2 - 1}$

$= 2 \int \left(1 + \frac{1}{u-1} - \frac{1}{u+1} \right) du$

$= 2 \left[u + \frac{1}{2} \ln(u-1) - \frac{1}{2} \ln(u+1) \right]$

$= 2\sqrt{x} + \ln(\sqrt{x}-1) - \ln(\sqrt{x}+1) + c$

c)

i) $y = \frac{4}{3}x, y = -\frac{4}{3}x$

ii) $\frac{2x}{9} - \frac{2y}{16} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{16x}{9y}$

$= \frac{48 \sec \theta}{36 \tan \theta}$

$= \frac{4 \sec \theta}{3 \tan \theta} \left(\frac{4}{3 \sin \theta} \right)$

iii)

$y - 4 \tan \theta = \frac{4}{3 \sin \theta} (x - 3 \sec \theta)$

$3 \sin \theta y - 12 \sin \theta \tan \theta = 4x - 12 \sec \theta$

$4x = 3 \sin \theta y - 12 [\sin \theta \tan \theta - \sec \theta]$

$4x = 3 \sin \theta y - 12 \left[\frac{\sin^2 \theta}{\cos \theta} - \frac{1}{\cos \theta} \right]$

$4x = 3 \sin \theta y + 12 \cos \theta$

iv) Now $4x = -3y$

$\therefore -3y = 3 \sin \theta y + 12 \cos \theta$

$(3 \sin \theta + 3)y = -12 \cos \theta$

$y = \frac{-12 \cos \theta}{3 \sin \theta + 3}$

$\therefore x = -\frac{3}{4} \left(\frac{-12 \cos \theta}{3 \sin \theta + 3} \right)$

$= \frac{3 \cos \theta}{\sin \theta + 3}$

$$\begin{aligned}
 \text{v) } \frac{3 \sec \theta}{2} &= \frac{3 \cos \theta}{1 + \sin \theta} \quad \checkmark \\
 \frac{3}{2 \cos \theta} &= \frac{3 \cos \theta}{1 + \sin \theta} \\
 3 + 3 \sin \theta &= 6 \cos^2 \theta \\
 3 + 3 \sin \theta &= 6 - 6 \sin^2 \theta \\
 6 \sin^2 \theta + 3 \sin \theta - 3 &= 0 \\
 3(2 \sin^2 \theta - 1)(\sin \theta + 1) &= 0 \\
 \sin \theta &= 1/2 \text{ or } -1 \\
 \sin \theta &\neq -1 \quad \left(\frac{3 \cos \theta}{1 + \sin \theta} \right) \\
 \therefore \sin \theta &= 1/2 \\
 \theta &= \pi/6 \quad \checkmark
 \end{aligned}$$

Question 3

$$\begin{aligned}
 \text{a) } \int \tan^3 x \, dx &= \int (\sec^2 x - 1) \tan x \, dx \quad \checkmark \\
 &= \int \sec^2 x \tan x \, dx - \int \tan x \, dx \quad \checkmark \\
 &= \frac{\tan^2 x}{2} + \ln(\cos x) + c
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int_0^{\pi/2} \sqrt{2 - 2 \cos 2x} \, dx \\
 &= \sqrt{2} \int_0^{\pi/2} \sqrt{1 - (1 - 2 \sin^2 x)} \, dx \quad \checkmark \\
 &= \sqrt{2} \int_0^{\pi/2} \sin x \, dx \\
 &= -2 [\cos x]_0^{\pi/2} \\
 &= -2 [0 - 1] \\
 &= 2. \quad \checkmark
 \end{aligned}$$

$$\text{c) i) } \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta \quad \checkmark$$

$$\text{ii) Let } x = \cos \theta$$

$$\therefore 4x^3 - 3x = \frac{\sqrt{3}}{2} \quad \checkmark$$

$$\text{if } 8x^3 - 6x - \sqrt{3} = 0$$

$$\therefore \cos 3\theta = \frac{\sqrt{3}}{2} \quad \checkmark$$

$$3\theta = \pi/6, 11\pi/6, 13\pi/6, \dots$$

$$\theta = \pi/18, 11\pi/18, 13\pi/18 \quad \checkmark$$

$$\therefore \text{roots } \cos \pi/18, \cos 11\pi/18, \cos 13\pi/18$$

$$\begin{aligned}
 \text{d) i) } x \rightarrow \infty \quad x^5 \rightarrow \infty \\
 x \rightarrow -\infty \quad x^5 \rightarrow -\infty \quad \checkmark \\
 \therefore \text{at least 1 real root}
 \end{aligned}$$

$$\text{ii) } P(x) = x^5 - 5cx + 1$$

$$P'(x) = 5x^4 - 5c$$

$$\text{if } c < 0 \quad \checkmark$$

$$P'(x) = 5x^4 + 5R \quad R > 0$$

For stationary pt.

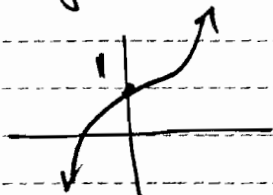
$$P'(x) = 0$$

$$\therefore 5x^4 + 5R = 0 \quad \checkmark$$

which has no real roots

$\therefore P(x)$ has only one real root.

$$\text{iii) } P(0) = 1 \quad \therefore \text{root is negative}$$



iv) For ³ distinct roots

∴ 2 stationary points ✓

$$\therefore P'(x) = 5x^4 - 5c$$

$$= 5(x^4 - c)$$

$$= 5(x^2 - c^{1/2})(x^2 + c^{1/2})$$

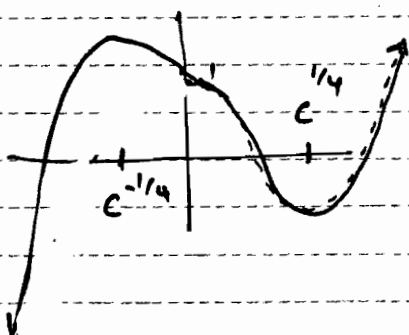
$$= 5(x^2 + c^{1/2})(x - c^{1/4})(x + c^{1/4})$$

$$P'(x) = 0 \therefore$$

∴ 2 stationary pt when

$$x = c^{1/4} \text{ or } -c^{1/4} \quad \checkmark$$

Now when $x = c^{1/4}$ $P(x) < 0$



$$\therefore (c^{1/4})^5 - 5c \cdot c^{1/4} + 1 < 0$$

$$c^{5/4} - 5c^{5/4} + 1 < 0$$

$$-4c^{5/4} < -1$$

$$c^{5/4} > 1/4 \quad \checkmark$$

$$c > (1/4)^{4/5}$$